

Phase 2

Requirement Analysis Phase

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1. Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux (supports cross-platform operations).
- **Node.js (v16 or above):** Runtime ecosystem running server logic and routing infrastructure.
- **npm (v8 or above):** Node package manager to control operational dependencies.
- **Gemini AI SDK:** Integration using @google/generative-ai for natural language insights.
- **MongoDB:** Document NoSQL engine storing users, posts, categories, comments, and analytics metrics.
- **Postman:** API verification toolkit to assert schema validation responses across administrative route shields.
- **Code Editor:** Visual Studio Code or similar IDE environment.

2. Hardware Requirements

- **Processor:** Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.
- **RAM:** 8 GB minimum (16 GB recommended for concurrent instances of MongoDB, processing layers, and execution nodes).
- **Storage:** 1 GB of available disk workspace.